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Attachment and leader's development via experiences☆

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ABSTRACT

Apart from biographies and memoirs of outstanding leaders, studies on the development of leaders in everyday life from childhood to maturity are rare. We propose a developmental model of leaders in which certain variables formed in early childhood are the basic components for leadership development in later life. These are a low level of trait anxiety [Spielberger, C.D. (1972). Conceptual and methodological issues in anxiety research. In C.D. Spielberger (Ed.), Anxiety: Current trends in theory and research. New York: Academic Press, vol. 1] and openness to experiences [Costa, P.T. Jr., & McCrae, R.R. (1992a). Revised NEO Personality Inventory (NEO-PI-R) and NEO Five-Factor Inventory (NEO-FFI) professional manual. Odessa, FL: Psychological Assessment Resources]. We argue that secure attachment style [Bowlby, J. (1969). Attachment and loss: Vol. 1. Attachment. New York: Basic Books] explains the formation and influence of these basic components, which in turn affect the leader's development through experience in leadership roles. Questionnaires were administered to 286 subjects. The data obtained were analyzed by structural equation modeling (SEM).The findings show that trait anxiety and openness to experience affect leaders' development via experiences. Alternative models in which formative variables may affect the magnitude and intensity of leadership experiences are discussed.

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A review of more than a century's studies on psychological aspects of leadership reveals a notable absence of comprehensive and methodical discussion and research on developmental psychological aspects of leaders (Avolio, 1999, 2005; Avolio & Gibbons, 1988; Day, Zaccaro, & Halpin, 2004; Klonsky, 1983; Kuhnert & Lewis, 1987; London, 2002; McCauley, & Van Velsor, 2004; Popper & Mayseless, 2002; Zacharatos, Barling, & Kelloway, 2000).

Most evidence on the development of leaders has been supplied by historians who studied the lives of outstanding leaders (e.g., Bullock, 1991; Burns, 1956). Characteristically, these studies (1) focus on exceptional leaders who are of great interest to historians, and (2) analyze their development mainly from a psychoanalytic viewpoint (Burns, 1978). Such studies are clearly limited with regard to the possibility of generalization. This criticism is often directed at psychoanalytical research in general (Meadow, 1996), but it is even more relevant in the case of historical studies of outstanding (usually political) personages (Burns, 1978). They do not necessarily shed light on leaders of local communities, organizational units, social frameworks, and the like, namely those who might be described as "leaders in everyday life."

A more comprehensive analysis of leaders' development is to be found in the works of scholars who discuss psychological processes that may generate patterns of development toward leadership (Aberbach, 1995; Kets de Vries, 1989; Popper, 2000; Zaleznik, 1992). These authors identified in some leaders similar types of socialization in early childhood, which, they claim, may indicate distinct patterns of development of leadership. For instance, the biographies of several leaders reveal a close relationship with the mother (sometimes as the mother's favorite child), expectations of greatness projected by at least one of the parents, and exposure to esteemed authority figures in the family or in social circles close to the family (Burns, 1978).

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Gibbons (1986) and Avolio and Gibbons (1988) examined the developmental process of leaders through in-depth personal interviews conducted with a sample of senior managers, who were asked to talk about events and experiences in their past from childhood to maturity. These authors enumerated seven factors in leaders' development, which can be clustered into three groups of variables: 1. Family conditions in which the individual was raised; 2. Experiences of leadership; 3. Openness to experiences.

Although Avolio and Gibbons' work contributes conceptually and empirically to the study of leaders' development, their conclusions are limited since their work is based on interviewing sixteen transformational leaders; without a comparative perspective, possible unique features in leaders' development cannot be seen. These can only be identified and examined through comparison with a population of "non-leaders". Moreover, the different factors in Avolio and Gibbons' study do not reflect the same theoretical entity. While openness to experiences is a personality variable, family conditions and leadership experiences represent environmental and situational variables.

Recently these limitations served as a point of departure in the planning of a series of studies comparing a group of leaders with a group of non-leaders on a battery of personality traits and developmental variables (Amit, Popper, Gal, Maman, & Lisak, 2008; Popper, Amit, Gal, Sinai, & Lisak, 2004). Leaders were found to be distinctly higher than non-leaders in self-confidence as measured by level of trait anxiety (Popper et al., 2004). (Trait anxiety, according to Spielberger (1972), differs from state anxiety in that it is a personality characteristic.) These findings are congruent with Judge, Bono, Ilies, and Gerhardt's (2002) findings on a negative correlation between neuroticism and leadership emergence (−.24). Other studies found that leaders (more than non-leaders) reported experiences of leadership roles at school and in social settings (Amit et al., 2008; Hall, Lord, Ritter, Swee, & Dubois, 2004; Maman, 2005). Leaders also proved higher than non-leaders in openness to experiences as measured by the NEO Personality Inventory of the Five-Factor model (Costa & McCrae, 1992a; Judge & Bono, 2000; Judge et al., 2002; Popper et al., 2004).

The difference between leaders and non-leaders in willingness to undertake leadership roles is congruent with the findings of studies on general development. For example, a correlation of .43 was found between openness to experiences and experience seeking (Zuckerman, Kuhlman, Joireman, Teta, & Kraft, 1993); a high correlation (.42) was found between openness to experiences and flexibility (McCrae, Costa, & Piedmont, 1993), and a negative correlation (−.56) was found between openness to experience and tough-mindedness (Conn & Rieke, 1994). The variable openness to experiences may also contribute to the understanding of the social dimension that is so relevant to close leadership. Guttman (1995) found that problems in interpersonal relationships were associated more with individuals who were classified as being less open to experiences. Similar findings were reported by Kitron (1976) and Gilbert (1991).

Popper et al. (2004) and Amit et al. (2008) showed that level of trait anxiety and openness to experiences carry the greatest weight in predicting leadership potential and leadership development in regression equations. These findings indicate the existence of basic components in the development of leaders — personality components formed in early childhood (Popper & Mayseless, 2002, 2007 that influence the scope and intensity of leadership-learning processes generated by leadership experiences (Amit et al., 2008; Maman, 2005).

The paucity of theoretical and empirical material on leaders' development from early childhood is surprising in view of the substantial theoretical and methodological progress made in psychological knowledge, particularly in understanding the individual's inner world as shaped in early childhood. The monumental contributions of Freud (1920), Erikson (1959), Jung (1986), Kohut (1971), Bowlby (1969), and many others have advanced our understanding of the growth of internal representations that guide the individual's emotions and behaviors throughout his/her life. Empirical research in developmental psychology has likewise made enormous progress in characterization of the personality and its development from childhood to old age (Bowlby, 1969; Cassidy, 1994; Hazan & Shaver, 1987). Yet, this knowledge has not found expression in the empirical study of leadership psychology (Popper, 2005; Popper & Mayseless, 2002).

This assumption that has been the basis for many if not most studies in developmental psychology (Schaffer, 2000) may well apply also to the development of individuals as leaders. Findings on the central importance of level of trait anxiety and openness to experiences in a comparison of leaders and non-leaders can provide a sound basis for examining the influence of these basic personality components, which are formed in early childhood and may be hypothesized as exercising a considerable impact on the learning of leadership in maturity. This appears to be a promising direction, first because of the important part played by variables formed in early childhood on the adult's development, and secondly because it may be inferred from studies conducted specifically on openness to experiences. For example, Costa and McCrae (1992b) showed that openness to experience as measured in early childhood remains stable over time. They reported a correlation of .66 between two measures of openness over a period of 24 years. Finn (1986) reported a correlation of .62 on intellectual interest across a gap of 30 years. Costa and McCrae (1992b) reported that despite the stereotype that adults are characterized by more conservatism and rigidity, analysis of the means of openness to experiences indicated a slight and non-significant decrease at a more advanced age.

Accordingly, differences between leaders and non-leaders in levels of trait anxiety and openness to experiences seem especially important from the point of view of development. Data on leaders' more extensive experience of leadership roles in childhood (Amit et al., 2008; Maman, 2005; Popper et al., 2004) may be explained by the leaders' lower level of trait anxiety and higher level of openness to experiences. That is, the basic capacity for learning and development that germinates in early childhood meets the opportunity — the relevant social context for the manifestation of leadership.

The inevitable question is this: What determines the low level of trait anxiety and the openness to experiences which, in our opinion, form the psychological substructure for the development of leadership via experience of leadership roles?

Bowlby's (1969) attachment theory may provide an answer to this question because (1) it deals with early childhood, (2) it permits operationalization of variables discussed at the conceptual level in psychodynamic theories, (3) it is considered today the most important developmental theory (Cassidy & Shaver, 1999), (4) it is based on empirical support of greater scope than any other

developmental theory (Cassidy & Shaver, 1999), and (5) some promising initial attempts have been made to link attachment theory with leadership (Davidovitz, Mikulincer, Shaver, Iszak & Popper, 2007; Mikulincer & Florian, 1995; Popper, Mayseless, & Castelnovo, 2000). We now examine the theory and its implications for the present discussion.

1. Overview of attachment theory

Based on ethological, evolutionary, and control systems concepts, attachment theory postulates an innate bio-social behavioral system in the infant, the purpose of which is to maintain proximity between the infant and its primary caregiver. The evolutionary purpose of this behavioral system is to promote the infant's survival through protection by a "stronger and wiser" figure, given that it cannot survive on its own.

According to attachment theory, adults have a complementary behavioral system, whose main function is protection of the infant. This system is activated when threat impends. The child uses the caregiver as a secure base from which to explore, and periodically checks to ascertain the availability of the attachment figure. When a threat arises (e.g., hunger, loud noises) the attachment system is activated and the child seeks the protection of the caregiver, who then serves as a safe haven.

Differences in the ability to signal the need and desire for closeness, and especially differences in the caregiver's responsiveness, generate variations in babies' attachment styles. It is assumed that from infancy onward children form an internal working model, which includes internalization and representations of major aspects of their attachment relationships with their caregivers (Bowlby, 1969, 1973; Bretherton, 1985). These aspects include perception of the self as (un)worthy of love and attention, and perception of the attachment figure as (un)willing and (un)able to attend to the attachment needs when they arise. The internal working model guides the interpretation and planning of interpersonal transactions with the caregiver, and later with other important figures. At first it involves a representation of the specific relationship with a specific caregiver. Later, following repeated experiences, the model increasingly becomes a part of the child's developing personality and turns into a more abstract, generalized representation of the self and others (Bretherton, 1985; Collins & Read, 1990).

On the basis of Bowlby's theory, Ainsworth, Blehar, Waters, and Wall (1978) identified three styles of infant attachment, one secure, and two insecure, namely ambivalent and avoidant. These styles were later also identified in childhood and adulthood (see reviews by Solomon & George, 1999). As summarized in a review chapter by Cassidy (1999), the internal working model of secure individuals includes a basic trust in their caregiver and confidence that he or she will be available, responsive, and helpful when needed. With this assurance, they are bold in their explorations of the world, and able both to rely on themselves and to turn to others when in need. In contrast, the internal working model of the anxious/ambivalent pattern is characterized by uncertainty as to whether the parent or caregiver will be available, responsive, or helpful when called upon. Avoidant attachment is a pattern in which individuals have no confidence that they will receive care when they seek it; on the contrary, they expect to be rebuffed. Thus, secure individuals have more chances of becoming self-confident and open than do insecure ones.

Many studies bear witness to the stability of these internal models over the individual's lifetime (Cassidy, 1994), and the relation between the inner world formed in infancy (attachment style) and feelings, behaviors and relationships that reach expression in maturity (e.g., Cicchetti, Toth, & Lynch, 1995; Mikulincer & Orbach, 1995). Research in the early 1990s addressed both the long-term stability of attachment patterns and the role that sensitive care plays in the formation of attachment bonds. Presenting initial findings from the Minnesota longitudinal project, Elicker, Englund, and Sroufe (1992) reported that assessments of attachment during infancy successfully predicted specific patterns of social behavior in adolescence, just as Bowlby had anticipated. Specifically, they affirmed a significant relationship between secure attachment as assessed in infancy and leadership ratings by observers at a weekend camp organized in adolescence, some 15 years later (Englund, Levy, Hyson, & Sroufe, 2000). Similar conclusions were reported by correlational research (e.g., Isabella, 1993; Isabella & Belsky, 1991). A great deal of research using Hazan and Shaver's self-report measures has tested and confirmed fundamental predictions of attachment theory as formulated by Bowlby (Collins & Read, 1990; Feeney, Noller, & Callan, 1994; Simpson, Rholes & Nelligan, 1992). Hazan and Shaver (1987) found that people with a secure attachment style were more capable of intimacy and closeness in romantic relations than those with an insecure style. Similar findings were reported by Mayseless, Sharabani, and Sagi (1997) in connection with parent–child relations and friendship. Recently, research based on attachment theory has been expanded to the spheres of work orientations (Hazan & Shaver, 1990), and particularly leadership in organizations (e.g., Keller, 2003; Keller & Cacioppe, 2001). The researchers' assumption was that leadership also involves relationships between the leader and his/her followers, so a theory predicting the quality of relationships can also be relevant to leadership.

Popper and Mayseless (2007) argue that in terms of attachment theory, leaders adopt the role of stronger and wiser caregivers and provide a safe haven and a secure base for their followers.

Descriptions of effective leaders in everyday life in the scientific literature (e.g., House & Howell, 1992; Howell, 1988; Shamir, House, & Arthur, 1993) closely resemble descriptions of security-promoting attachment figures. Effective leaders are sensitive and responsive to their followers' needs; they provide group members with advice, guidance, and emotional and instrumental resources; they support their followers' creativity, initiative, and autonomy; they enhance their self-worth and self-efficacy; they support their followers' desire to meet new challenges and acquire new skills. These leaders can be viewed as sensitive and responsive caregivers who provide their followers with a sense of security and a platform for personal growth and development (Mayseless & Popper, 2007). Such leaders must have the ego resources required for taking on these kinds of leadership roles. The argument is that secure individuals have the psychological base required for such leadership whereas insecure personality types lack it. Mikulincer and Florian (1995) found that individuals defined as having a secure attachment style were assessed as more suitable for officer roles in sociometric evaluations by their peers on a military course. Popper et al. (2000) found a significant positive correlation between transformational leadership and a secure attachment style.

In sum, attachment theory appears to supply the integrative theoretical framework for characterizing the development of the basic components described in the introduction. Since attachment styles predict abilities in the affective social domain, as well as curiosity and daring (Cassidy & Shaver, 1999), presumably they evince operational expression in level of trait anxiety and openness to experiences. Specifically, the argument here is that attachment styles, by definition and nature, can predict levels of trait anxiety and openness to experiences, which, as stated, are the basic components for the later development of leadership via experiences. The research question, then, is this: Does a secure attachment style predict low trait anxiety and high openness to experiences, which are the psychological substructure necessary for learning leadership via experience? And if so, to what extent does trait anxiety level and openness to experiences predict leadership experience?

This argument can be presented through a general causal model suggesting that secure attachment style influences trait anxiety level and openness to experiences, and these two variables in turn influence the dependent variable. We tested the argument using structural equation modeling (SEM). Following Hoyle's (1995) recommendations for a simplified presentation of the SEM model, we show in Fig. 1 the general conceptual causal model.

The two preliminary hypotheses in this study relate to an argument represented by lines H1 and H2 in the general model (Fig. 1).

H1. A negative correlation will be found between secure attachment style and a high level of trait anxiety.

H2. A positive correlation will be found between secure attachment style and openness to experiences.

2. Influence of the basic components on the development of leaders via experience

Substantiation of these hypotheses would lead to examination of the influence of these relations on the future development of leaders. The general hypothesis is that trait anxiety and openness to experiences are variables affecting the number and scope of leadership experiences. It is rooted in the argument (elaborated below) that leadership experiences are a major route in the development of leaders (Avolio, 1999, 2005; Avolio & Gibbons, 1988; Chan & Drasgow, 2001; Day, 2000; Day et al., 2004; London, 2002; McCall, Lombardo & Morrison, 1988). All major theories of learning and developmental psychological processes explicitly or implicitly place experience at the center of the learning process. For example, the learning theories generally identified with Skinner (1989) are based on the law of effect, that is, behavior is guided by past results. If in the past a certain behavior was rewarded, thereby receiving positive reinforcement, the probability of that behavior appearing in the future is higher, while behaviors that were not rewarded will probably appear less and less, and may even disappear.

Aspects such as self-efficacy also develop on the basis of experiences. Bandura (1977, 1986), for example, showed that the sense of success based on experiences in certain areas led to the strengthening of self-efficacy in those areas. These principles proved relevant and applicable to the area of leadership development. For example, Akin (1987), Kotter (1988), and McCall et al. (1988) reported that managers retrospectively viewed practical experience as a cardinal learning experience in their development as leaders. Successful experiences in leadership roles, whether in the family framework or in educational or social frameworks, show the individual that he or she is perceived by others as a leader; they also strengthen the individual's own belief in his or her ability to be a leader. Leaders' testimonies reveal that personal experience was an important element in their learning of leadership (Chan & Drasgow, 2001; Day, Sin, & Chen, 2004; Kotter, 1988, 1990). Atwater, Dionne, Avolio, Camobreco, and Leu (1999), seeking predictors of leadership, examined a large number of mental and physical characteristics of freshmen at a military academy. They found that past leadership experiences and self-efficacy differentiated levels of leadership most clearly: students with a rich background of leadership experiences and high self-efficacy belonged to the highest level of leadership. Similarly, Hall et al. (2004) found in a longitudinal study of leadership that West Point cadets who had had more leadership experiences at high school showed higher initial leadership performance at West Point. Lord and Hall (2005) argue that leadership development via experiences can be described in terms of qualitative changes in both process and knowledge as skills develop from the novice to the expert level (e.g., Ericson & Charness, 1994). These changes involve development of a large repertoire of more targeted, domain-specific productions rather than the general heuristics applied to all

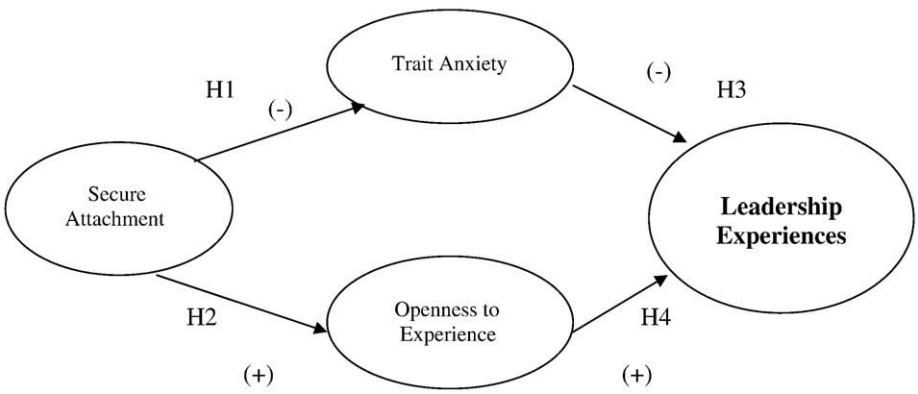


Fig. 1. The general conceptual causal research model.

superficially similar situations. In addition to guiding behaviors and social perceptions, this experience-based knowledge of leadership becomes over time inextricably integrated with the development of one's self-concept as a leader.

Similarly, Sternberg (1981) highlights the notion of *tacit knowledge* (TK) — a form of implicit knowledge that helps the individual to solve real-world, practical problems. To acquire tacit knowledge, however, an individual must have been exposed to the knowledge via experience. Experience-based knowledge facilitates performance by allowing individuals to define the novel problems they face in an environment functionally similar to the one in which they acquired the knowledge. Sternberg, Wagner, Williams, and Horvath (1995) found that TK distinguished more from less successful performers in several domains, including sales, bank management, and clerical work. TK was recently applied to research in the domain of military leadership (Hedlund et al., 2003). The results indicated that domain-specific TK could explain individual differences in leadership experiences. In fact, Cianciolo, Antonakis, and Sternberg (2004) suggest that an experience can be used as a "mentor", fostering the creation of meaningful new knowledge based on past experiences.

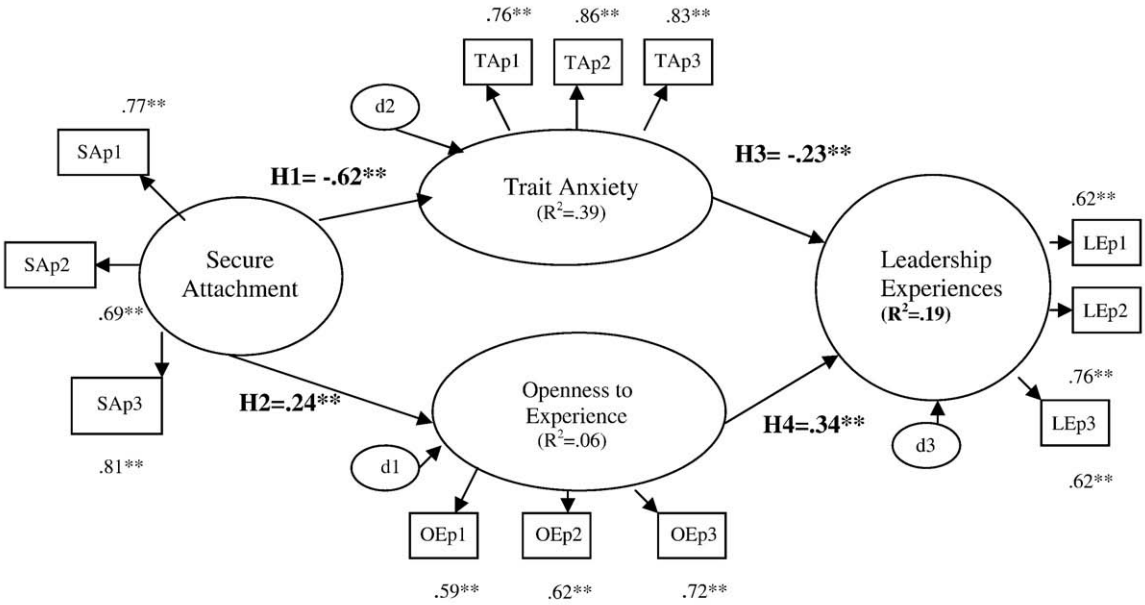
In sum, leadership experiences, according to various scholars, are an important key to the development of both leadership skills and self-efficacy, as well as to the development of the cognitive and affective capacities vital for the successful functioning of a leader.

Our hypothesis is that the more an individual is characterized by a low level of trait anxiety and a higher level of openness to experiences, the more he or she will undergo experiences relevant to the development of leadership. This general hypothesis was examined specifically by two lines in the general model (Fig. 1), H3 and H4:

H3. A negative relation will be found between levels of trait anxiety and leadership experiences.

H4. A positive relation will be found between openness to experiences and leadership experiences.

Figs. 1 and 2 present our suggested model represented by hypotheses one to four. In addition, five alternative causal models were tested. Model 2 adds a direct line between secure attachment and leadership experience, and model 3 suggests that all three independent variables (secure attachment, trait anxiety and openness to experiences) are directly related to the dependent variable — leadership experience. We also tested the possibilities in which trait anxiety was the independent variable (model 4) and openness to



* p< 0.05 ** p< 0.01

d = error term
p = parcel, for each of the latent constructs (all variables) 3 parcels were randomly formed.

Model 1: Fit measures

x^2(df)	57.3 (50) p=0.22
NFI	.94
NNFI	.99
CFI	.99
ECVI	.48
RMSEA	.025*

Fig. 2. Leadership development pattern: research model 1 ($N = 238$).

experience was the independent variable (model 5). Finally, we tested an alternative model in which the independent variable consisted of trait anxiety and secure attachment (model 6). Although it is outside the theoretical claim, we also tested the possibility that leadership experiences in childhood may predict attachment style.

3. The research

3.1. Participants

The research took place in the Israel Defense Forces (IDF). The participants were 286 male soldiers (who formed a representative sample of the entire cohort in compulsory army service) toward the end of their basic training. They belonged to four companies: two from the armored corps (169 soldiers) and two from the infantry (117). The soldiers volunteered to participate and anonymity was assured. Self-report questionnaires were administered to the participants. The response from the armored companies was 65% and from the infantry 55%. Between 10% and 20% of the soldiers in the various platoons were new immigrants who had language difficulties answering the questionnaires and were excluded from the sample. Other soldiers who did not participate were absent for technical reasons (on duty, sick leave, etc.), and it may be assumed that they were omitted at random. Prior to the research, a pilot study was conducted on 195 recruits in the armored corps and the infantry; that study served as the basis for examination of the questionnaires.[1]

3.2. Research instruments

3.2.1. Attachment style questionnaire

Attachment styles among adults are measured in research in two ways (Bartholomew & Shaver, 1998), one based on a clinical interview and the other on quantitative measures developed consequent to Hazan and Shaver's pioneering work (1987) on attachment and romantic relations. Hazan and Shaver's work stemmed from the assumption that chronically lonely people who constantly seek romantic relations and regularly fail in them in fact reflect a late expression of insecure attachment styles in early childhood. Hazan and Shaver developed a self-report questionnaire based on the three attachment styles of Ainsworth et al. (1978). In responding to the questionnaire, people were asked to think of romantic situations in their past and to indicate which statements, referring to the three styles, were the most descriptive of themselves.

In subsequent studies this measure and several variants have been used for a host of close relationships. Some studies have correlated this measure with retrospective reports of childhood experiences with parents. The bulk of research in this tradition has focused on the influence of attachment patterns on personal adjustment and adult relationships (for reviews see Rholes & Simpson, 2004).

Hazan and Shaver (1987) showed that the distribution of attachment styles of the adults in their study was similar to that found among the infants (Ainsworth et al., 1978), with 56% reporting a secure attachment style, 24% an avoidant style, and 20% an anxious/avoidant style. A similar distribution was found in various other studies conducted among adults; also, as described above, attachment style was found to predict other affective relations that are measurable among adults (e.g., Collins & Read, 1990; Feeney et al., 1994; Mikulincer & Orbach, 1995). These studies validated statistically the argument that attachment styles formed in early childhood can indeed be reflected by measurement in a questionnaire for adults.

In recent years there has been some criticism of the instrument developed by Hazan and Shaver (1987). The general argument is that the typology of three supposedly distinct types cannot encompass the variety and complexity of the attachment schemas. Factor analyses performed in various studies yielded two mega-dimensions that help to explain individual differences in attachment: avoidance and anxiety (e.g., Fraley & Waller, 1998; Shaver, Belsky, & Brennan, 2000). The anxiety dimension is characterized by constant worry about relationships, fear of loneliness, and fear of abandonment. The avoidance dimension is characterized by discomfort with closeness and dependence, avoidance of expression of emotion, and extreme self-reliance. Low scores on anxiety and avoidance characterize the secure style. This measurement method is the most widely used today in research on individual differences in attachment tendencies, and this was the approach we used in the present study. We calculated the secure score on the 13 items characterizing the anxious (6 items) and the avoidant (7 items), so those who were neither anxious nor avoidant were secure (a typical item measuring attachment anxiety: “I worry about being abandoned.” A typical avoidant attachment item: “I prefer not to show a partner how I feel deep down.”)

3.2.2. Trait anxiety inventory

This inventory presents 20 statements that reflect the respondent's general anxiety level (Spielberger, Gorsuch, & Lushene, 1970). The respondent is asked to rate the degree to which he or she generally feels each emotion on a four-point scale from “almost never” to “almost always” (typical items: “I often get tired”, “I am often sad”). After reversal of the items, the scores on the questionnaire range between 20 and 80, the higher score indicating a higher level of trait anxiety. The test–retest reliability of the

[1] To test the various instruments used in the research we conducted a pilot study, from February to the end of October 2002, with the participation of 195 soldiers from one armored corps company. First a series of self-report questionnaires was administered to the soldiers, who were approaching the end of their basic training (approximately two months in the army). At the second stage we returned two weeks later to 28 of the soldiers (approx. 14%) for a semi-structured in-depth interview. At the third stage (May 2002) we went back to the soldiers yet again, and administered the questionnaires to test their stability.

questionnaire, as reported in the manual, is .73. Internal consistency reliability tests yielded Cronbach's alpha values between .89 and .91 (Teichmann & Melinek, 1979).

3.2.3. Questionnaire on openness to new experiences

To examine the degree of openness to experiences, the research team developed a questionnaire based on the NEO-PI-R Personality Inventory (Costa & McCrae, 1992a). It examines five personality factors (the Big Five), one of which is openness to new experiences. In the Hebrew version of the questionnaire this factor is examined by 48 items (Montag & Levin, 1994). The respondent is asked to indicate how far he or she agrees with each statement (typical statements: "I often think differently from others", "I like surprises"). Replies are given on a five-point Likert scale from "strongly disagree" to "strongly agree." Cronbach's alpha reliability coefficient on the "openness to experience" factor in this Hebrew version was .65. Our research team developed an abbreviated questionnaire, *Openness to Experiences* (OE), comprising 24 items. The main reasons for constructing the new questionnaire were the need to shorten the original one due to the limited time available for administering it in the army setting, and the need to modify the language to make it more comprehensible to the soldiers. The general score on this measure can range from 24 to 120.

The new questionnaire (OE) was validated against the Hebrew version of the original questionnaire (NEO-PI-R) by administration of both questionnaires to 113 respondents, men and women, most of them (65%) students at Haifa University, the rest attending preparatory courses (for admission to the university). Cronbach's alpha reliability coefficient of the new questionnaire was .64, and a significant correlation ($r = .77$, $p < .01$) was found between the scores of the original and the new questionnaire. The latter was examined in the context of the pilot study, and its reliability level was found acceptable and matching the level reported in the literature (see footnote 1).

3.2.4. Leadership development questionnaire

To examine the development of leadership we used a questionnaire that we constructed to examine leadership-shaping experiences (LSE) from childhood to conscription in the military. The questionnaire was composed on the basis of Avolio and Gibbons' (1988) measure, and incorporated other issues that proved central in semi-structured interviews conducted in the pilot study (see footnote 1).

The logic behind the construction of this measure was the possibility of quantifying issues previously investigated by qualitative methods (Amit et al., 2008). Leadership experiences was measured by seven questions (on a five-point Likert scale) indicating the extent of responsibility and influence (i.e., roles, duties, tasks) respondents experienced at home, at school, and in social frameworks.

The questions were: To what extent did you feel during your childhood that your parents or others around you encouraged you to influence others? To what extent were you given responsibility at home (taking care of younger children, household chores, etc.)? At junior high school and high school, to what extent did you have roles that included certain responsibilities (class committee, students' council)? To what extent did you like to undertake organizational tasks at school? To what extent did you try to change things in your school? To what extent did you have any experience of leadership roles in other social frameworks? How would you define the experience you accumulated? (1 = hardly any experience–5 = rich experience).

3.3. Analysis method

The reliability of the research instruments was tested and the relations between the research variables were examined by means of a correlation matrix. We used Structural Equation Modeling with AMOS 5 software (Byrne, 2001) to test the hypothesized models. Following Anderson and Garbing's (1988) *two-step* approach, we tested the measurement model first, and then the structural causal model.

The measurement model assesses whether all items on a given scale represent the same latent construct. A rule of thumb is that a measurement model with three indicators per latent construct is ideal, and it can carry up to five indicators without estimation difficulties (Bagozzi & Heatherton, 1994). When more than five items measure the same construct, a parceling procedure can be used to combine items randomly into composites (Bagozzi & Heatherton, 1994). Parceling reduces random errors and simplifies the model while simultaneously maintaining the integrity of multiple-indicator measurement. We used parceling procedures for all the variables in the models. Each of our variable measures had more than five items. In each measure these were randomly parceled into three composite indicators, which were entered into the measurement and structural models as multiple indicators to estimate the latent constructs.

The structural model provides maximum likelihood estimates of all identified model parameters, and evaluates the degree to which the model reproduces the observed variance–covariance matrix in terms of a chi-square goodness-of-fit statistic (Bollen & Long, 1993). This method enabled us to test the fit of the correlation matrix against alternative models. Our research model (model 1) suggests that the secure attachment style influences the trait anxiety level and openness to experiences level; these two variables, in turn, affect the dependent leadership variable. This research model is based on the assumption that certain psychological phenomena expressed in later development arise from the internal representations formed during attachment in early infancy. This assumption was supported, for example, in the longitudinal Minnesota studies (Elicker et al., 1992) and in cross-sectional studies on attachment (Englund et al., 2000; Isabella, 1993; Isabella & Belsky, 1991). This is the theoretical basis for numerous studies on the prediction of emotional phenomena in adults (see, for example, Rholes & Simpson, 2004). The SEM model, which serves a confirmatory analysis for theory testing, is thus an adequate method for testing this theoretical claim. In addition to the main model, we have tested all other alternative models mentioned earlier.

4. Results

The reliability scores of the research instruments and the correlation coefficients of the variables (constructs) under study are presented in Table 1. The instrument's reliability scores were sufficient, as compared to the relevant scores reported in the literature. Correlations between the variables went in the directions hypothesized: significant negative correlations between trait anxiety, openness to experiences, secure attachment style and leadership experiences; significant positive correlations between all the other variables: secure attachment style, openness to experiences, and leadership experiences.

4.1. Measurement model

The results from our measurement model indicate that the model fitted the data well. A non-significant χ^2 (51.97; $df=48$; $p=.32$) was obtained, indicating that the model fitted the data. Other fit statistics were excellent (CFI = .996, NFI = .95, NNFI = .994, GFI = .96, RMSA = .02). Factor loadings ranged from 0.59 to 0.86, in support of the model's measurement properties, and correlations between the constructs ranged from .21 to .61 and were significant. All these measures indicated good model fit, with satisfactory convergent and discriminant validity. Hence, we could proceed to the next step.

4.2. Structural model

Six structural models were tested and the results are presented in Figs. 2–7, following Hoyle's (1995) recommendations. For each model the results are presented in diagrams with specification of the main statistical measures: regression weights, multiple correlation coefficients, and fit measures of each structural model.

Our hypothesized model, model 1, fitted the data well, as shown in the fit measures: a non-significant χ^2 was obtained (57.3; $df=50$; $p=.22$), indicating that the model fitted the data, all fit indexes exceeded .90, and RMSEA was significant ($p<.05$).

Fig. 2 presents our structural model graphically, making it possible to test the research hypotheses. According to H1, a negative correlation would be found between secure attachment style and high level of trait anxiety. The findings fully supported this hypothesis, with a significant ($p<.01$) and strong negative relationship between secure attachment and trait anxiety ($\beta=-.62$). H2 posited that a positive relation would be found between secure attachment style and openness to experiences. The findings supported this hypothesis, with a significant ($p<.01$) positive relationship between secure attachment and openness to experiences ($\beta=.24$). According to H3, a negative relation would be found between levels of trait anxiety and leadership experiences. This hypothesis was supported, with a significant ($p<.01$) negative relation between trait anxiety and leadership experiences ($\beta=-.23$). The findings also supported the next hypothesis, H4: a positive relation would be found between openness to experiences and leadership experiences. A significant ($p<.01$) positive relation was indeed found here ($\beta=.34$).

Model 1, depicted in Fig. 2, support our hypothesis, showing that secure attachment style directly affects the levels of trait anxiety and openness to experiences, and these two basic components affect leadership experiences. This model explained 19% of the variance of the leadership parameter.

As mentioned, other alternative models were tested. In model 2, a direct line was added between secure attachment and leadership experiences. As presented in Fig. 3, the model fitted the data well: a non-significant χ^2 was obtained (54.77; $df=49$; $p=.27$), all fit indexes exceeded .90, and the RMSEA was significant ($p<.05$). However, the additional direct relation between secure attachment and leadership experiences was not significant ($\beta=.18$, $p=.12$). Moreover, the relation between trait anxiety and leadership experiences was not significant ($\beta=-.11$, $p>.05$).

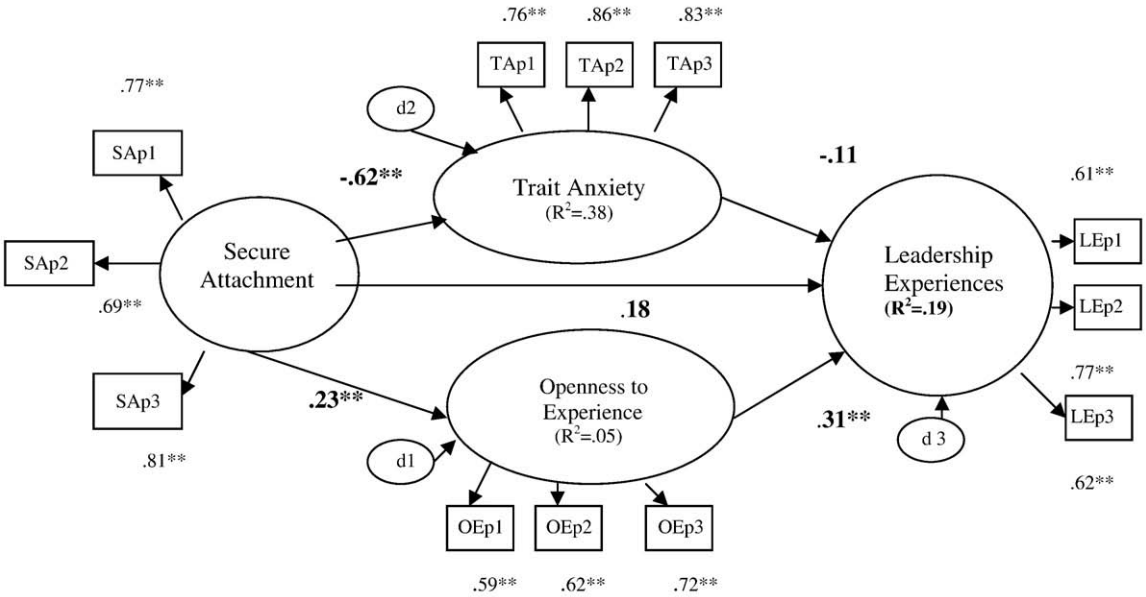
To test the differences between the two models a chi-square difference test was conducted and was found not significant ($\chi^2=2.36$, $df=1$, $p=.12$). These findings suggest that this alternative model made no significant contribution and was less adequate in explaining the relations between the variables.

Another alternative model suggested that three independent variables (secure attachment, trait anxiety and openness to experiences) were directly related to the dependent variable (model 3). As presented in Fig. 4, the model did not fit the data well: a significant χ^2 was obtained (132.1; $df=51$; $p=.00$), most fit indexes did not exceed .90, and the RMSEA was not significant ($p>.05$). In this model the relationship between trait anxiety and leadership experiences was not significant. We tried to improve this model by adding correlations between all three independent variables, but although the fit of the model was improved only the relation between openness to experiences and leadership experiences was found significant. These findings suggest that this alternative model is not adequate.

Table 1
Internal consistency reliability (Cronbach's alpha) of the research instruments and correlations between the variables.

Questionnaire	No. of items (no. of cases*)	Alpha	Mean (STD)	1.	2.	3.
1. Secure attachment style	13 (245)	.80	4.98 (.841)			
2. Trait anxiety	20 (229)	.88	36.65 (9.53)	−.53**		
3. Openness to experience	24 (239)	.68	85.79 (8.55)	.18**	−.21**	
4. Leadership experiences	7 (240)	.70	2.63 (.69)	.25**	−.24**	.26**

**Significance $p<.01$; *Significance $p<.05$.



* p< 0.05 ** p< 0.01

d = error term
p = parcel, for each of the latent constructs (all variables) 3 parcels were randomly formed.

Model 2: Fit measures	
x^2(df)	54.77 (49) p=0.27
NFI	.94
NNFI	.99
CFI	.99
ECVI	.48
RMSEA	.022*

Fig. 3. Leadership development pattern: research model 2 ($N = 238$).

In model 4, anxiety is the main independent variable predicting the secure attachment style and openness to experiences, and these two variables in turn predict leadership experiences.

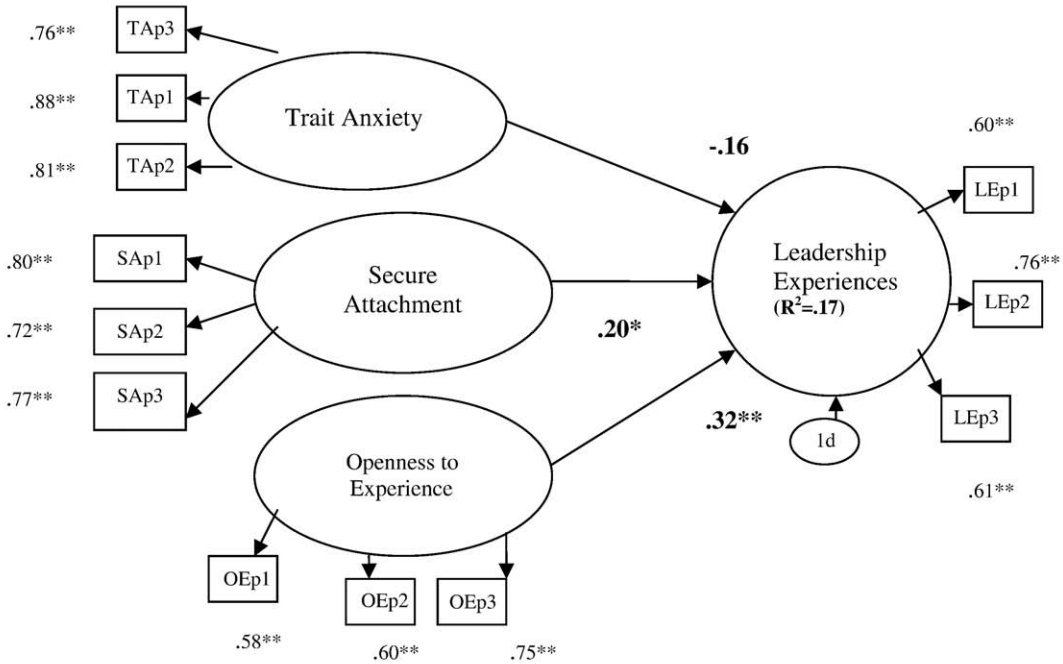
Model 4 fitted the data well: a non-significant χ^2 was obtained (52.55; $df = 49$; $p = .34$), all fit indexes exceeded .90, and the RMSEA was significant ($p < .05$). However, two direct relations proved not significant: the relation between anxiety and leadership experiences ($\beta = -.08$, $p > .05$) and the relation between secure attachment and leadership experiences ($\beta = .21$, $p = .07$). Still, the fit of this alternative model indicates that anxiety is negatively and significantly related to secure attachment style.

To test whether openness to experience may be the main independent variable predicting attachment and anxiety we tested another alternative, model 5, presented in Fig. 6.

Model 5 did not fit the data as well as model 4: a significant χ^2 was obtained (113.5; $df = 49$; $p = .00$), some fit indexes did not exceed .90, and the RMSEA was not significant ($p > .05$). The relationship between trait anxiety and leadership experiences ($\beta = -.11$, $p > .05$), as well as that between attachment and leadership experiences ($\beta = .17$, $p = .06$), were not significant.

To test the possibility that leadership experiences in childhood and adolescence predict attachment style as well as anxiety and openness to experience, another model was examined. This one did not fit the data well: a significant χ^2 was obtained (108.25; $df = 51$; $p = .00$), most fit indexes did not exceed .90 (NNFI = .88, NFI = .92, RFI = .85, CFI = .93), the RMSEA was not significant ($p > .05$), and ECVI score was .69.

Comparing ECVI scores for each of the alternative models indicated that the three best fitting were models 1, 2 and 4 (χ^2 differences between each pair of models cannot be checked since the models are not nested). These three models had the lowest ECVI scores (since model 2 was found not significantly different from model 1, we refer only to model 1). Model 1, our suggested model, shows that secure attachment is the main independent variable, whereas according to model 4 anxiety is the main independent variable. This finding may indicate that the relation between secure attachment and anxiety could represent a mutual causal effect. We tried to test such a model using a causal arrow between anxiety and secure attachment going in either direction (retaining all other variables in the model), but the model obtained was non-recursive with a stability index of 1.842. According to



* p< 0.05 ** p< 0.01

d = error term
p = parcel, for each of the latent constructs (all variables) 3 parcels were randomly formed.

Model 3: Fit measures

x^2(df)	132.1 (51) p=0.00
NFI	.86
NNFI	.88
CFI	.91
ECVI	.79
RMSEA	.082

Fig. 4. Leadership development pattern: research model 3 ($N = 238$).

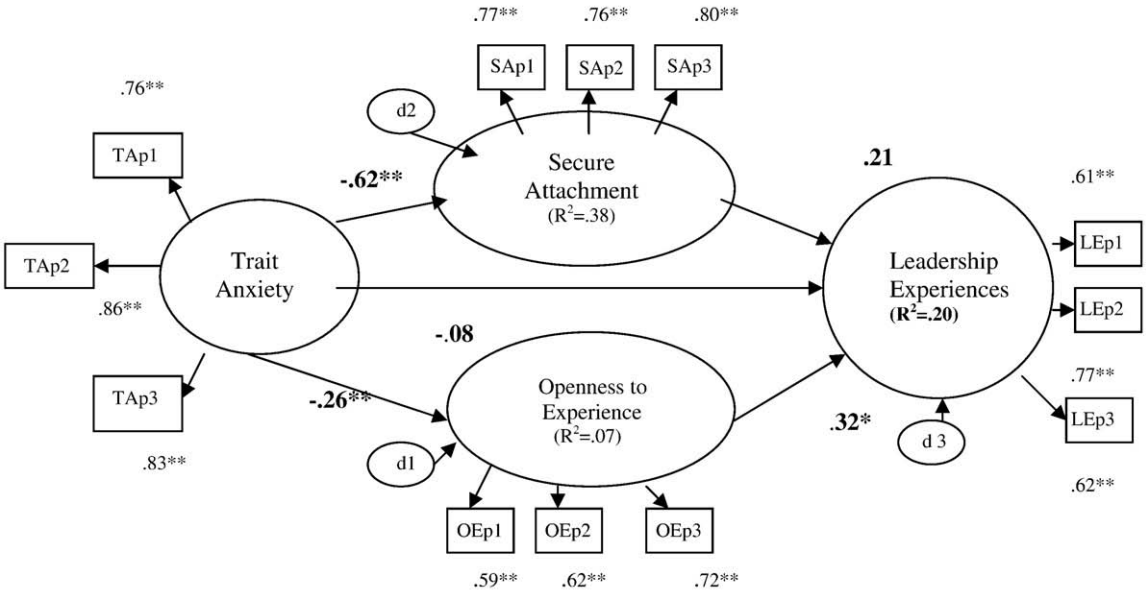
Bentler and Freeman (1983), such an unstable model presents problems of interpretation. A stable and fitted model was obtained when we changed the causal arrow between secure attachment and anxiety into a correlation; this is presented in the last model (model 6) in Fig. 7. According to this model both secure attachment and low level of anxiety, predict openness to experience and explain leadership experiences.

5. Discussion

This study was based on a hierarchical assumption that a psychological substructure formed in early childhood gives rise to the basic components that later influence the intensity and scope of development of leadership via experiences. This argument was tested by structural equation modeling (SEM), a procedure that facilitates assessment of the validity of the measures in the model as well as the hypothesized causal path.

The study was designed to examine as a preliminary hypothesis our assumption of the formation of a psychological substructure for the basic components. Attachment theory (Bowlby, 1969) – and secure attachment style as a measurable expression of processes described in this theory – served as the theoretical framework for prediction of the formation of the basic components, trait anxiety and openness to experiences, in early childhood.

Six alternative models assumed to predict leadership experiences were presented and tested (the possibility that leadership experiences is the independent variable was also tested and ruled out). The results support our original hypothesis (model 1) showing that secure attachment style directly affects the levels of trait anxiety and openness to experiences, and these two basic components are related to leadership experiences. This model explained 19% of the variance of the leadership parameter. The direct relation between secure attachment and leadership experiences added in model 2 was not significant ($\beta = .18$, $p = .12$). Nor



* p< 0.05 ** p< 0.01

d = error term
p = parcel, for each of the latent constructs (all variables) 3 parcels were randomly formed.

Model 4: Fit measures

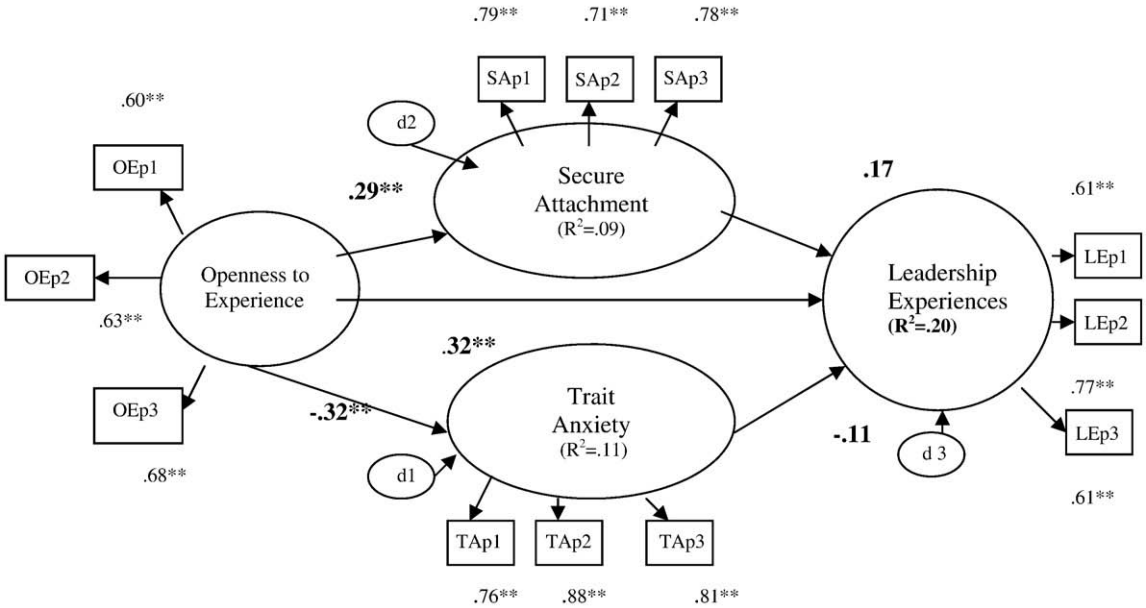
x^2(df)	52.55 (49) p=0.34
NFI	.96
NNFI	.95
CFI	.99
ECVI	.47
RMSEA	.018*

Fig. 5. Leadership development pattern: research model 4 ($N = 236$).

was the relation between trait anxiety and leadership experiences. Model 3, in which secure attachment, trait anxiety and openness to experiences were directly related to the dependent variable was found inadequate. Model 5, in which openness to experience was the independent variable, was also found inadequate. However, trait anxiety (presented in model 4) proved significant in predicting secure attachment and openness to experience. That is, two of the tested models can be seen as competing: the original model, in which the independent variable is secure attachment style, or the model in which trait anxiety is the basis of the causal model. Model six suggests not seeing these two variables in terms of causal relationships but seeing both of them as the infrastructure of the entire model leading to prediction of the dependent variable .This perspective is consistent with competing views regarding the formation of general ("trait") anxiety. One claims that anxiety is mostly learned (Izard, 1977). This approach allows more psychological developmental explanations (such as attachment theory) in which anxiety is formed as a result of certain forms of interactions occurring mostly in early childhood. This approach is largely supported by clinical psychologists (see reviews by Spielberger, 1979; Barlow, 1988).

The other approach deals with anxiety in terms of genetic, psychophysiological and neurobiological aspects (Selye, 1976) or biological theories of personality emphasizing different levels or intensities of biological variables such as cortical arousal (e.g. Eysenck, 1981) or other heritable tendencies such as an active behavioral inhibition system (e.g. Cloninger 1986).

The results in this study cannot determine whether anxiety is more learned or more innate. However, the negative significant correlation found between trait anxiety and secure attachment style (Figs. 2, 3, 5 and 7) can be meaningful in an analysis of the development of central emotional capacities which have been widely studied in attachment research and are important in maintaining effective leadership. For example, as mentioned in the review, it was found that a person who is secure has a better capacity for intimacy in romantic relations (Hazan & Shaver, 1987) and a better quality of friendship relations (Mayseless et al., 1997). These findings are congruent with studies indicating a positive correlation between secure attachment style and the developing factor in the factorial structure of transformational leadership — *individualized consideration* (Popper et al., 2000; Davidovitz et al., 2007). Common to all these findings is the development of affective capacity, which, as Kellet, Humphrey, and Sleeth (2006) assert, is central to (close) leader–followers relationships. The ability to identify others' emotions and the ability to



* p< 0.05 ** p< 0.01

d = error term
p = parcel, for each of the latent constructs (all variables) 3 parcels were randomly formed.

Model 5: Fit measures

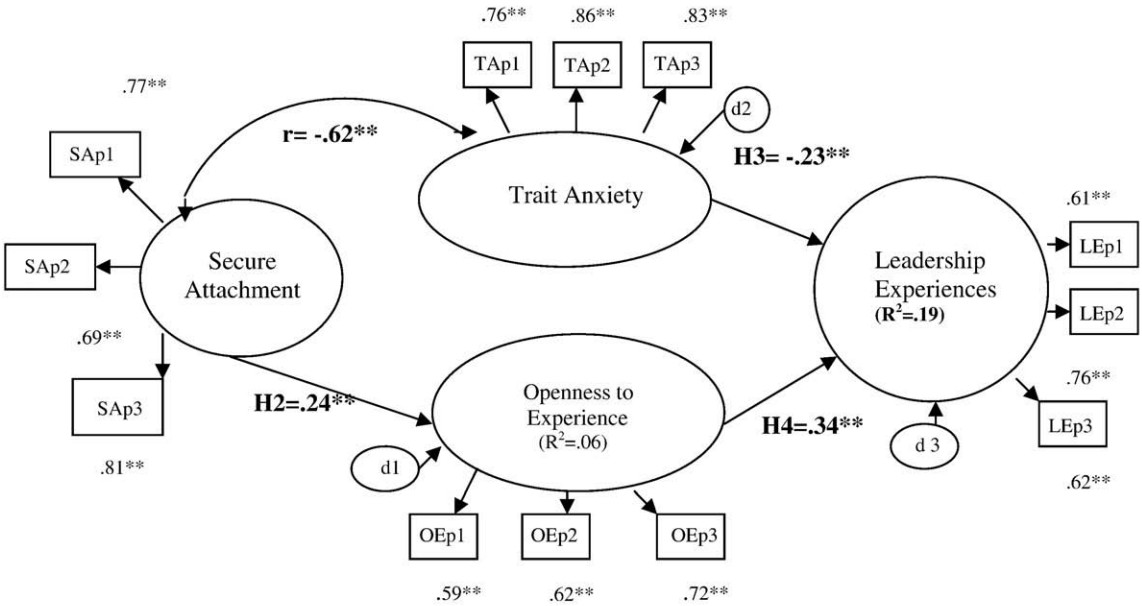
x^2(df)	113.50 (49) p=0.00
NFI	.90
NNFI	.88
CFI	.93
ECVI	.73
RMSEA	.075

Fig. 6. Leadership development pattern: research model 5 ($N = 236$).

display one's own emotions are reported as crucial in developing effective leader–member relations (Newcomb & Ashkanasy, 2002). Of particular importance to such relationships is the ability to extend empathy, that is, "the ability to comprehend another's feelings and to re-experience them oneself" (Salovey & Mayer, 1990, pp. 194–195). Goleman, Boyatzis, and McKee (2002, p. 50) state that empathy is "the fundamental competence of social awareness." They argue that effective leaders create a "resonance" with their followers, and by doing so they can guide their followers to more productive emotional responses and work behavior. So emotional qualities that are necessary for the existence of social relations of any kind seem also relevant to leadership relations. The secure attachment style seems to predict the functions that are characterized by a higher quality of interaction with other people in a variety of relationships. Our findings strengthen this theoretical position and support what appears to be a promising direction in looking at leadership as a certain type of emotional relationship (Popper, 2004).

The negative significant correlations found between trait anxiety and openness to experience (Figs. 5 and 6) might also add information that can be interpreted in developmental terms by using the attachment theory perspective. It is reported in attachment studies that a person can afford to explore the environment and new experiences because of the general sense of a secure base (Ainsworth et al., 1978) and also because of a high sense of efficacy (Carnelley & Ruscher, 2000; Cassidy, 1999; Hazan & Shaver, 1990). These findings are congruent with reports on correlations between openness to experiences and leadership, using the Big Five model (Costa & McCrea, 1992a; Judge et al., 2002).

All our research findings indicate on the centrality of trait anxiety and openness to experience to leaders' development via experiences. Although more exploration is needed on whether these variables are more acquired (e.g Bowlby, 1969) or more innate (Cloninger, 1986), the findings still indicate that individuals are likely equipped with differential developmental substructures that might affect leadership development. Moreover, the attempt made in this study to link conceptualizations and studies in developmental psychology and the study of leadership development is far from exhausted. Future studies should also deal with the issue of motivation to lead, an aspect that has been neglected in the psychological study of leadership (Chan & Drasgow, 2001); only recently have some attempts been made to deal with it (Kark & Van Dijk, 2007; Popper & Mayseless, 2002). So far these attempts have been mostly theoretical, and much more empirical work needs to be done. The assumption is that not all of those



* p< 0.05 ** p< 0.01

d = error term
p = parcel, for each of the latent constructs (all variables) 3 parcels were randomly formed.

Model 6: Fit measures

x^2(df)	57.31 (50) p=0.22
NFI	.94
NNFI	.99
CFI	.99
ECVI	.48
RMSEA	.025*

Fig. 7. Leadership development pattern: research model 6 ($N = 236$).

who have basic developmental components for leadership will necessarily want to be leaders, just as people who have abilities in any other area may not want to engage in it and therefore will not endeavor to develop those abilities. This naturally has developmental implications.

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